

AWS Cloud Curriculum — Services & Tools

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1 DevOps & AWS Learning Hub

A structured path from zero to advanced, plus two handbook series:

Resource	Target size	Status
DevOps Curriculum	Reference list	Complete
AWS Curriculum	Reference list	Complete
DevOps Handbook	~1,200 pages	TOC + Part I-II written; Parts III-XII outlined
AWS Handbook	~800 pages	TOC + Part I-II written; Parts III-X outlined

1.1 Download PDFs

Pre-built PDFs are in the [pdf/](#) folder:

PDF	Description
devops-handbook.pdf	DevOps handbook — 11 chapters + full TOC
aws-handbook.pdf	AWS handbook — 11 chapters + full TOC
devops-curriculum.pdf	DevOps tools & concepts reference
aws-curriculum.pdf	AWS services reference
devops-aws-complete-guide.pdf	All content in one PDF

Rebuild PDFs after editing markdown:

```
./scripts/build-pdfs.sh
```

Requires `pandoc` and `wkhtmltopdf`.

- 1. **Start with the curricula** — they list every tool, concept, and AWS service in learning order.
- 2. **Follow the handbook TOCs** — each chapter has a page budget; work through sequentially.
- 3. **Expand on demand** — ask an AI assistant or instructor to flesh out any chapter number from the TOC.

1.2 Suggested study pace

Phase	Duration (self-paced)	Focus
Foundations	4-8 weeks	Linux, Git, networking, scripting
Core DevOps	8-12 weeks	CI/CD, containers, IaC, cloud basics
Intermediate	8-12 weeks	Kubernetes, observability, security
Advanced	12+ weeks	SRE, platform engineering, multi-

Advanced	12+ weeks	engineering, multi-cloud, FinOps
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1.3 Page convention

Each handbook “page” $\approx$ 400 words of instructional content + diagrams/commands. Labs and exercises add extra pages in the TOC estimates.

2 AWS Cloud Curriculum: Services & Tools (Beginner → Advanced)

All major AWS services grouped by domain, with learning priority.

Legend: ★ = core (learn first) · ◆ = intermediate · ◇ = advanced/specialized

2.1 1. Account, Identity & Governance ★

Service / Tool	Purpose
AWS Organizations ★	Multi-account structure, SCPs
IAM ★	Users, groups, roles, policies, MFA
IAM Identity Center (SSO) ★	Workforce federation, permission sets
AWS Control Tower ◆	Landing zone automation
AWS Config ◆	Resource compliance & history
AWS CloudTrail ★	API audit logging
AWS Service Catalog ◇	Governed self-service provisioning
AWS Resource Access Manager (RAM) ◆	Cross-account resource sharing
AWS Account Factory ◆	Automated account vending (Control Tower)
AWS CLI / CloudShell ★	Command-line & browser shell
AWS Management Console ★	Web UI
AWS CloudFormation ★	Infrastructure as Code (native)
AWS CDK ◆	IaC in TypeScript/Python/etc.
Terraform + AWS Provider ★	Industry-standard IaC
AWS Tagging & Resource Groups ★	Cost & ops organization

2.1.1 Concepts

- Shared responsibility model
 - Root account hygiene (MFA, no daily use)
 - Least privilege & permission boundaries
 - SCP vs IAM policy
 - Cross-account roles & assume-role
 - AWS global vs regional services
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2.2 2. Networking & Content Delivery ★

Service	Purpose
VPC ★	Virtual network, subnets, route tables
Internet Gateway / NAT Gateway ★	Public & private egress
VPC Endpoints (Gateway & Interface) ◆	Private AWS API access

Security Groups & NACLs ★	Stateful vs stateless firewalls
Elastic IP ★	Static public IPs
Route 53 ★	DNS, health checks, routing policies
CloudFront ★	CDN, edge caching
ALB / NLB / GWLB ★	Layer 7, 4, and gateway load balancers
VPC Peering / Transit Gateway ◆	Inter-VPC & hybrid connectivity
AWS VPN / Direct Connect ◆	On-prem hybrid networking
PrivateLink ◆	Private service exposure
AWS Global Accelerator ◇	Anycast for global apps
Route 53 Resolver ◆	Hybrid DNS
AWS Network Firewall ◇	Managed perimeter firewall
VPC Lattice ◇	Service-to-service networking

2.2.1 Tools

- **Reachability Analyzer, VPC Flow Logs, Traffic Mirroring**

2.3 3. Compute ★

Service	Purpose
EC2 ★	Virtual machines, AMIs, instance types
Auto Scaling Groups ★	Horizontal scaling
Elastic Beanstalk ◆	PaaS for web apps
Lambda ★	Serverless functions
ECS / Fargate ★	Container orchestration (AWS-native)
EKS ★	Managed Kubernetes
AWS Batch ◆	Batch/HPC jobs
Lightsail ◆	Simplified VPS
App Runner ◆	Container PaaS
EC2 Image Builder ◆	Automated AMI pipelines
AWS Outposts / Wavelength / Local Zones ◇	Edge & hybrid compute
Serverless Application Repository ◇	Share Lambda apps

2.3.1 Tools

- **AWS Systems Manager** (Session Manager, Run Command, Patch Manager)
- **EC2 Instance Connect**

2.4 4. Storage ★

Service	Purpose
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S3 ★	Object storage, versioning, lifecycle
EBS ★	Block storage for EC2
EFS ◆	Managed NFS
FSx ◆	Windows/Lustre/ONTAP/OpenZFS file systems
S3 Glacier ◆	Archive tiers
Storage Gateway ◇	Hybrid storage
AWS Backup ◆	Centralized backup
DataSync / Transfer Family ◆	Large data migration

2.4.1 Concepts

- Storage classes (S3 Standard, IA, Glacier)
- Encryption at rest (SSE-S3, SSE-KMS, SSE-C)
- S3 bucket policies vs ACLs
- EBS volume types (gp3, io2, st1)

2.5 5. Databases ★

Service	Purpose
RDS ★	Managed MySQL, PostgreSQL, MariaDB, Oracle, SQL Server
Aurora ★	AWS-native relational, serverless option
DynamoDB ★	Serverless NoSQL key-value/document
ElastiCache ◆	Redis / Memcached
DocumentDB ◆	MongoDB-compatible
Neptune ◇	Graph database
Timestream ◇	Time-series
Keyspaces ◇	Cassandra-compatible
Redshift ◆	Data warehouse
QLDB ◇	Ledger database
MemoryDB for Redis ◆	Durable Redis
RDS Proxy ◆	Connection pooling

2.6 6. Application Integration & Messaging ★

Service	Purpose
SQS ★	Managed message queues
SNS ★	Pub/sub notifications
EventBridge ★	Event bus, scheduling
Step Functions ◆	Workflow orchestration
MQ ◆	Managed ActiveMQ/RabbitMQ

AppSync ♦	GraphQL API
API Gateway ★	REST/WebSocket/HTTP APIs

2.7 7. Developer Tools & CI/CD ★

Service	Purpose
CodeCommit ♦	Git repos (legacy; many use GitHub)
CodeBuild ★	Managed build
CodeDeploy ♦	Deployment automation
CodePipeline ★	CI/CD orchestration
CodeArtifact ♦	Package repository
Cloud9 ♦	Cloud IDE
CDK / SAM / Copilot ♦	IaC & serverless/container tooling
X-Ray ♦	Distributed tracing
CloudWatch Synthetics ♦	Canary monitoring

2.8 8. Observability & Management ★

Service	Purpose
CloudWatch ★	Metrics, alarms, dashboards, logs
CloudWatch Logs Insights ★	Log query
AWS X-Ray ♦	Tracing
Systems Manager ★	Ops hub (Parameter Store, Patch, Session)
CloudTrail ★	Audit
AWS Health Dashboard ★	Service events
Trusted Advisor ♦	Best-practice checks
Personal Health Dashboard ★	Account-specific events
Service Quotas ♦	Limit management
OpsWorks ◇	Legacy Chef/Puppet (avoid for new projects)

2.9 9. Security, Identity & Compliance ★

Service	Purpose
IAM ★	(see section 1)
KMS ★	Encryption key management
Secrets Manager ★	Rotating secrets
Certificate Manager (ACM) ★	TLS certificates

WAF & Shield ♦	Web firewall & DDoS
GuardDuty ★	Threat detection
Inspector ♦	Vulnerability scanning
Macie ♦	S3 data classification
Security Hub ♦	Central security findings
Detective ◇	Security investigation
Firewall Manager ♦	Central WAF/rule management
Artifact ♦	Compliance reports
Audit Manager ◇	Continuous audit evidence
CloudHSM ◇	Dedicated HSM
Cognito ♦	User pools & identity for apps
Verified Permissions ◇	Cedar policy engine
IAM Access Analyzer ♦	External access findings

2.10 10. Migration & Transfer ♦

Service	Purpose
Migration Hub	Migration tracking
Application Migration Service (MGN)	Lift-and-shift
Database Migration Service (DMS)	DB replication
Server Migration Service	(legacy → MGN)
Transfer Family	SFTP/FTPS/FTP to S3
Snow Family	Physical data transfer appliances

2.11 11. Analytics & Big Data ♦

Service	Purpose
Athena ♦	SQL on S3
Glue ♦	ETL & data catalog
EMR ♦	Hadoop/Spark
Kinesis ♦	Streaming (Data Streams, Firehose, Analytics)
QuickSight ♦	BI dashboards
Lake Formation ♦	Data lake governance
MSK ♦	Managed Kafka
OpenSearch Service ♦	Search & analytics (Elasticsearch fork)
Data Pipeline ◇	Legacy orchestration
Clean Rooms ◇	Privacy-preserving analytics

2.12 12. Machine Learning & AI ◇

Service	Purpose
SageMaker ◇	End-to-end ML platform
Bedrock ◇	Foundation models API
Comprehend / Rekognition / Textract / Polly / Transcribe ◇	AI APIs
Forecast / Personalize / Kendra ◇	Specialized ML
CodeWhisperer / Q Developer ◆	AI-assisted development

2.13 13. IoT, Media & Specialized ◇

Service	Purpose
IoT Core / Greengrass	Device connectivity
MediaConvert / Live / Package	Video processing
GameLift	Game server hosting
WorkSpaces / AppStream	Virtual desktops
Connect	Contact center

2.14 14. Cost & Billing ★

Service / Tool	Purpose
Cost Explorer ★	Spend analysis
Budgets ★	Alerts & limits
Cost & Usage Report (CUR) ◆	Detailed billing export
Pricing Calculator ★	Estimate costs
Savings Plans / Reserved Instances ◆	Commitment discounts
Compute Optimizer ◆	Right-sizing recommendations
Billing Conductor ◇	Reseller/MSP billing

2.14.1 Concepts

- Pay-as-you-go vs reserved vs spot
- Tagging strategy for cost allocation
- FinOps culture & unit economics

2.15 15. Well-Architected Framework (conceptual ★)

Six pillars — learn alongside every service:

1. **Operational Excellence** — run & monitor systems
2. **Security** — protect data & assets

3. **Reliability** — recover from failure
4. **Performance Efficiency** — use resources well
5. **Cost Optimization** — avoid unnecessary spend
6. **Sustainability** — minimize environmental impact

Tool: AWS Well-Architected Tool (workload reviews)

2.16 Certification path (recommended order)

Order	Certification	Focus
1	Cloud Practitioner (CLF)	Broad AWS vocabulary
2	Solutions Architect Associate (SAA)	Design & core services
3	Developer Associate (DVA)	CI/CD, serverless, APIs
4	SysOps Administrator (SOA)	Operations & monitoring
5	DevOps Engineer Professional (DOP)	Advanced CI/CD & automation
6	Solutions Architect Professional (SAP)	Complex multi-account design
7	Security Specialty	Deep security services
8	Networking Specialty	VPC, hybrid, TGW

2.17 Hands-on lab progression

1. Create IAM user, MFA, CLI profile
2. VPC with public/private subnets + NAT
3. EC2 + ALB + Auto Scaling
4. S3 static site + CloudFront + Route 53
5. RDS Multi-AZ + Secrets Manager
6. Lambda + API Gateway + DynamoDB
7. ECS Fargate or EKS cluster with CI/CD
8. CloudWatch dashboards + alarms + X-Ray
9. Terraform modules for reusable infra
10. Multi-account landing zone (Organizations + Control Tower)